

# CBAM engine: formulas, routes, tooling and what changed

Technical reference · 8 August 2026 · Delta Climate Research

Every figure below was read from the pack live at `deltacclimate.earth` on the date of writing, not from a local build. Where something is unverified it says so.

## 1. What shipped

Four changes, across two repositories, all live.

| Commit               | Repo  | What                                                                                      |
|----------------------|-------|-------------------------------------------------------------------------------------------|
| <code>397457e</code> | CBM   | Production-year markers become validity windows; generator emits its own thresholds again |
| <code>95224d0</code> | CBM   | Real OJ hashes pinned for IR 2025/2620 and 2025/2621                                      |
| <code>d529b1b</code> | Angad | Re-vendor: 185 unpriceable goods become 23                                                |
| <code>bf211f1</code> | Angad | Re-vendor: the two regulations now carry real hashes                                      |

Effect on the calculator, measured across all 574 catalogue CN codes at five origins, import date in 2026:

|                                             | Before | After |
|---------------------------------------------|--------|-------|
| Goods that produce an estimate              | 385    | 547   |
| Goods that cannot be priced from any origin | 185    | 23    |

**No calculated value changed.** A migration gate proved the multiset of (CN code, column, benchmark value) identical before and after. The recovered goods were previously returning `status: 'unavailable'` — the engine refused rather than mispriced. This is coverage, not accuracy.

### Also this branch: multi-line, a real threshold verdict, and export

Six more changes, one repository (Angad only — no vendored file touched, and `cbam-sync-check.mjs` still reports the vendored engine intact).

| Commits              | Repo  | What                                                                                                                                                                                                               |
|----------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31c4ba4 – 8936429    | Angad | Line model, plus two SHA-256 digests: a per-line fingerprint of inputs as entered, and a pack-snapshot hash replacing the "browser-prototype" placeholder                                                          |
| 5a98e98 – fc7cf86    | Angad | Per-year threshold: lines grouped by calendar year and handed to <code>aggregateThresholdBasis / evaluateThreshold ; below_threshold</code> reachable only when the user ticks a per-year completeness attestation |
| d434711 – 9b2120c    | Angad | CSV export: one row per line, engine values verbatim, each figure beside its legal locator                                                                                                                         |
| 800df31 ,<br>9e8f641 | Angad | Renderers for the year-threshold card, running totals, per-line card and the printable audit document; the second commit fixed a defect that had printed a Commission workbook's hash under a regulation's label   |
| 516fb3a – bdb8162    | Angad | Wired <code>lines[]</code> into the page: add/remove lines, export buttons, print stylesheet                                                                                                                       |
| a700e22 ,<br>f04b95b | Angad | Post-ship fixes: Add-line failures now reported, focus restored on Remove, a stuck print class fixed, and an aria-live summary that had reintroduced the exact false certainty this branch exists to remove        |

The calculator went from pricing one line to pricing as many as the user adds, from a threshold verdict that could only ever say `indeterminate` to one that can say `below_threshold` on the user's own attestation, and from no export at all to a CSV and a printable audit document. Full detail in §4.3 and §4.4.

## 2. The formulas

Two regulations, two halves of one calculation.

**Certificates owed** are proportional to:

$$(\text{SEE} - \text{SEFA}) \times \text{mass} \times \text{CBAM factor} \times \text{certificate price}$$

`SEE` is what the good emitted. `SEFA` is the deduction that keeps imports symmetric with EU producers still receiving free ETS allowances. Omit `SEFA` and you overcharge by the entire free allocation.

### 2.1 Embedded emissions — Annex IV, Reg (EU) 2023/956

Simple good:  $\text{SEEG} = \text{AttrEmg} / \text{ALg}$   
 Complex good:  $\text{SEEG} = (\text{AttrEmg} + \sum m_i \cdot \text{SEE}_i) / \text{ALg}$

Our public calculator does not compute these. It reads the Commission's **default values** for the good, origin and production route, and applies the published mark-up. Computing `SEE` from plant data is the workspace product's job.

## 2.2 Free allocation — IR (EU) 2025/2620

$$\begin{aligned}
 \text{Eq 1 } \text{FAAg} &= \text{SEFAg}_{,y} \times \text{Mg} \\
 \text{Eq 2 } \text{SFAProcg} &= \text{CBAMy} \times \text{CSCFy} \times \text{BM*g} && \leftarrow \text{BM* is Column A} \\
 \text{Eq 3 } \text{SEFAg} &= \text{SFAProcg} && (\text{simple good}) \\
 \text{Eq 4 } \text{SEFAg} &= \text{SFAProcg} + \sum m_i \cdot \text{SEFA}_i && (\text{complex good}) \\
 \text{Eq 6 } \text{SEFAg} &= \text{CBAMy} \times \text{CSCFy} \times \text{BMg} && \leftarrow \text{BM is Column B}
 \end{aligned}$$

The calculator runs **Equation 6 exclusively**.

## 2.3 Column A versus Column B

The single most consequential decision in the engine.

| Column   | Covers                                       | Used when                                                                 |
|----------|----------------------------------------------|---------------------------------------------------------------------------|
| <b>A</b> | this installation's own process step only    | the emissions figure is process-only, with precursors declared separately |
| <b>B</b> | the whole finished good, precursors embedded | the emissions figure is all-inclusive                                     |

**The column is chosen by the SCOPE of the emissions figure, never by whether the data was verified.** An operator's verified actual data is normally all-inclusive and therefore still takes Column B.

Worked from our own rule pack — the cement chain:

| CN       | Good                      | Column A | Column B |
|----------|---------------------------|----------|----------|
| 25231000 | Cement clinker, route (A) | 0.666    | 0.666    |
| 25232900 | Portland cement           | 0        | 0.666    |
| 25232100 | White Portland cement     | 0        | 0.859    |

A clinker kiln calcines limestone, so its own process benchmark equals its whole-product benchmark. A cement grinder mills bought-in clinker and emits almost nothing itself, so its Column A is zero and its entire allocation arrives through the precursor term. The arithmetic closes exactly:

$$0 \text{ (process)} + 1 \text{ t} \times 0.666 \text{ (clinker's Column B)} = 0.666 = \text{cement's Column B}$$

**Column A plus Equation 4 is identical to Column B**, done properly. Get it wrong on a default value and you deduct zero where 0.666 was owed — roughly **€4,890 per 100 t** of CN 25232900. Across the shipped pack, 102 of 141 goods publishing both columns disagree.

The Column A path is implemented, audited, and **deliberately unreachable** from the public tool. It is only correct against a verified process-only figure with a declared precursor list, and a screening tool can obtain neither.

## 3. Production routes

### 3.1 The vocabulary

Both regulations publish the same closed legend. Anything outside it is a bug.

| Sector             |                           |                             |
|--------------------|---------------------------|-----------------------------|
| Cement             | (A) grey clinker / cement | (B) white clinker / cement  |
| Steel — carbon     | (C) BF/BOF                | (D) DRI/EAF · (E) Scrap/EAF |
| Steel — low alloy  | (F) BF/BOF                | (G) DRI/EAF · (H) Scrap/EAF |
| Steel — high alloy | (J) EAF                   |                             |
| Aluminium          | (K) primary               | (L) secondary               |

### 3.2 It is a matrix, not a list

The steel letters encode two orthogonal dimensions:

| Grade ↓ / Process → | BF/BOF | DRI/EAF | Scrap/EAF |
|---------------------|--------|---------|-----------|
| Carbon              | (C)    | (D)     | (E)       |
| Low alloy           | (F)    | (G)     | (H)       |
| High alloy          | —      | —       | (J)       |

This is not written down anywhere in the regulations; it falls out of the legend. It explains everything below.

### 3.3 How a route is selected

country of origin

└ IR 2025/2621 Annex I → "underlying production route determining CBAM BM"

└ IR 2025/2620 §5.3 → the benchmark row for that CN + route

The benchmark tables carry **no country column**. Country reaches the benchmark only through the route.

Two rules travel with it:

- "If no production route is indicated for a CN code, the CBAM benchmark is independent of the production route." (2621 Annex I)
- Unlisted country, or a — in the cell, falls back to the "Other countries and territories" table.

Both are implemented.

### 3.4 The defect fixed in 397457e

Annex §5.3 also publishes **production-year** variants: (1) for 2026-27 and (2) for 2028-30. These are validity periods, not routes.

`build-fa-package.py` was writing them into the `routeIndicator` field and giving every row one open-ended validity. Two consequences:

1. Both variants stayed active for every date, so the date could not choose.
2. Nothing ever matched them — the defaults corpus never declares a route of (1), nor a compound (F)(1).

**794 of 2,465 rows were unreachable.**

The fix moves the year into `validFrom / validTo`, where `resolveBenchmark` already looks — it filters `active(validFrom, validTo, date)` *before* matching the route. No engine code changed. The consumer was right; only the producer was wrong.

IR 2025/2621's own legend, read afterwards, lists (A) through (L) and contains no (1) or (2) — independent confirmation from the sister regulation.

Route distribution in the live pack, post-fix:

```
(none) 800  (C) 309  (D) 309  (E) 309
(F) 151  (G) 151  (H) 151  (J) 151
(K) 63  (L) 63  (A) 4  (B) 4
```

Zero year markers. (F) – (J) at 151 each are the former standalone rows plus their two former year-variants, now separated by window instead of by name.

### 3.5 Combined routes — known, not yet resolved

IR 2025/2621 Annex I publishes (C)/(F) **140 times** and (E)/(H) **25 times**. Read against the matrix, these are *process known, grade ambiguous* — the only combinations that exist are process-matched pairs. (C)/(D) and (D)/(G) never appear.

For every such CN the 2620 table publishes exactly **one** of the two halves, because the CN description settles the grade (§5.2.3: "*low alloy steel means alloy steel other than high-alloy*"). Checked across all 19 affected goods: one half present, zero ambiguous, zero missing.

`resolveBenchmark` does exact string equality, so (C)/(F) matches nothing and those goods fail closed. Splitting on / and taking the published half would recover 19 of the remaining 23. **Not yet implemented.**

## 4. The tool

### 4.1 Inputs

Good (CN code) · Country of origin · Production route · Emissions scope · Net mass (tonnes) · Import date.

The route list is never free-typed — it is derived from the routes the defaults corpus actually publishes for that CN and origin. The emissions-scope control appears only for goods the Commission publishes an indirect default for (cement, fertilisers, sintered iron ore); elsewhere it cannot change the answer.

### 4.2 What it returns

| Status                    | Meaning                                                            |
|---------------------------|--------------------------------------------------------------------|
| <code>cscf_pending</code> | A labelled what-if. No final figure exists.                        |
| <code>unavailable</code>  | Fails closed — names the missing rule selector verbatim.           |
| <code>zero_by_fiat</code> | Electricity: nil allocation by law (Art 1(2)), not by calculation. |

**94.3% of all resolvable answers are `cscf_pending`.** Zero are settled figures. The cross-sectoral correction factor is unpublished for 2026-2030 and is **not 1.0 by default**; the engine refuses to assume one and shows a labelled scenario at the last value the Commission actually set — `CSCF_2021_25 = 1 (sefa.ts)`.

**That scenario is a floor, not a midpoint.** The CSCF only ever scales the free allocation deduction (SEFA) down from its benchmark ceiling — it is bounded above by 1, never above it — so pinning it at 1 computes the *largest* deduction the law can produce and therefore the *smallest* certificate figure the good can owe. The real CSCF for 2026-2030, once published, can only be equal to or lower than 1. **A displayed figure cannot fall; it can only rise or hold.** Since 94.3% of answers are what-ifs, this governs almost every number the tool produces — an importer who reads a `cscf_pending` figure as a ceiling, rather than a floor, will under-provision.

### 4.3 The de minimis threshold

50 t per calendar year, across cement, aluminium, fertilisers and iron & steel (Reg 2023/956 Art 2(3)).

|                                     |                   |
|-------------------------------------|-------------------|
| above 50 t                          | → above_threshold |
| at or below + completeness=complete | → below_threshold |
| at or below + partial/unknown       | → indeterminate   |

**"Above" can be proven from one line. "Below" cannot** — it requires the whole year. The calculator now takes multiple lines and evaluates the threshold **per calendar year** through `aggregateThresholdBasis`; `below_threshold` is reachable only when the user explicitly attests the list is their complete year, and every surface that shows the verdict names that attestation as its basis. Estimates export as a CSV (engine values verbatim, one row per line, each beside its legal locator) and as a printable document whose final section states what the figures cannot tell you.

## 4.4 Export: CSV and the printable document

**CSV.** One row per line, engine values verbatim — no locale formatting, no rounding. The identity that reconciles a row is not "chargeable equals embedded minus free allocation" — that fails on two counts the regulation itself requires:

$$\text{chargeable\_tco2e} = \max(0, \text{direct\_tco2e} - \text{free\_allocation\_tco2e}) + \text{indirect\_tco2e}$$

Free allocation is a direct-emission benchmark (SEFA, §2.2), so it is deducted from the direct side only — indirect is added back **after** the deduction, not folded into what gets deducted from. And the deduction floors at zero: the regulation surrenders certificates, it never issues them, so a clean producer whose free allocation exceeds its own direct emissions cannot generate a negative charge.

Both halves are reachable with ordinary catalogue goods, checked against the live pack:

| CN / origin / route            | direct | free allocation | indirect | chargeable | What it shows                                                                                                    |
|--------------------------------|--------|-----------------|----------|------------|------------------------------------------------------------------------------------------------------------------|
| 25232900 / AL / default, 100 t | 99.0   | 64.935          | 3.3      | 37.365     | Deduction clamps nothing; indirect still adds on top                                                             |
| 25070080 / AO / (A), 100 t     | 24.2   | 64.935          | 4.4      | 4.4        | Free allocation exceeds direct emissions; the deduction floors at 0 and the whole bill is the indirect component |

**Printable document.** Four sections — what you asked, what we computed, on what authority, and what this does not tell you. The last section is the differentiator: it states the CSCF is unpublished and every figure a floor (§4.2), that Art 9 carbon-price deductions are not modelled, that any below-threshold verdict rests on the user's own completeness statement, and that the line fingerprint covers inputs as typed, never a source document.

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## 5. Data and provenance

Live pack, generatedAt 2026-08-07T16:59:36.563Z :

| Table           | Rows                                                       |
|-----------------|------------------------------------------------------------|
| defaultFactors  | 41,100                                                     |
| benchmarks      | 2,465 (Column A 661, Column B 1,804)                       |
| classifications | 574                                                        |
| cbamFactors     | 9 — 2026 is <b>0.975</b> , free allocation <i>retained</i> |
| cscf            | 5 — all pending , 2026-2030                                |
| prices          | 4 — 2026-Q1 €75.36, Q2 €75.28, Q3/Q4 unpublished           |
| thresholds      | 1 — 50 t                                                   |

7.55 MB raw, **281 KB on the wire** (Vercel serves brotli). Repeat visits transfer nothing.

## 5.1 Source hashes

Two regulations now carry real hashes, replacing sixty-four zeros:

```
IR 2025/2620 8bbba79e7f33f0e4943140c28e91a8810612f2fa770bd6dcad33fdb7045e4c05
IR 2025/2621 3155016c2e07b049b64f1ac4c2320061534245b81971ce5cba7814736f09acb4
```

Both PDFs were retrieved by hand and verified as genuine Publications Office artefacts. EUR-Lex sits behind an AWS WAF challenge, so neither can be re-fetched by a script — the hash pins *which* text we read; it cannot be re-verified automatically.

Four sources still carry placeholders: `dir-2003-87-art-10a-1a` , `dr-2019-331-art-14-6` , `reg-2023-956` , `ec-certificate-price-page` .

## 5.2 The second defect, fixed in the same commit

`build-fa-package.py` could not reproduce its own golden file. The `thresholds` block — the 50 t gate — and its `reg-2023-956` source had been hand-added on 2026-07-29 and never taught to the generator. **Every regeneration silently deleted the rule deciding whether an importer owes anything at all.**

Invisible until something regenerated. The year-window work forced a regeneration, which is how it surfaced.

The near-miss is worth recording: the first fidelity check compared `benchmarks` only, and passed. A full-package comparison found **two** drifts — `thresholds` *and* the `sources` array. **A fidelity check must compare the whole artefact; comparing the part you are about to change proves nothing about the parts you are not.**



### 5.3 Three digests, three different claims

The export (§4.4) and this document both now carry real hashes where the tool used to carry a placeholder — but they are not one claim wearing three names, and conflating them was a real defect caught in review before this branch shipped:

| Digest                                                                    | Over                                                                   | Says                                                                                              |
|---------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| IR 2025/2620, IR 2025/2621 (§5.1)                                         | the enacted regulation PDFs themselves                                 | this is the binding legal text we read                                                            |
| Commission benchmarks / default-values workbooks                          | the Commission's own Excel workbooks                                   | this is the informational transcription the defaults corpus was built from — not the binding text |
| <code>pack_snapshot</code> (CSV column) / <code>stamp.snapshotHash</code> | <code>generatedAt</code> plus both workbook hashes, SHA-256'd together | this is the exact corpus a given figure was computed from                                         |

```
Benchmarks workbook
b79108b025e697822f0f59de477fa68066c1c05c228fae2270cd230af84e8a7b
Default-values workbook
865372ed23649b7b02c9124f207fc0b0875fd244c45c19e9fb8cdb1e503a5003
```

An earlier build of the printable document read the workbook hashes but labelled them "IR (EU) 2025/2620" and "IR (EU) 2025/2621" — every character of each hash was real, but printed under the wrong artefact's name, which is a false provenance claim regardless. Fixed before ship ( `9e8f641` ): the document now prints the regulations' own hashes under those labels and the two workbook hashes separately, labelled as transcriptions. `stamp.snapshotHash` no longer reads "browser-prototype" ; every estimate is decorated with the real pack-snapshot hash before it renders.

## 6. How it is verified

- Unmodified generators reproduce their committed goldens **byte-for-byte**, before any change is made.
- A migration gate asserts every benchmark value, every `sourceLocator` and every non-benchmark section is unchanged.
- A regression test asserts no year marker can reappear in `routeIndicator`, and no (CN, column, route) group holds both a bounded and an unbounded row.
- `cbam-sync-check.mjs` hashes the eleven vendored engine files against a committed manifest. The engine in the website is a copy of the SaaS's, and this is the tripwire against drift.
- CBM: 379/379 tests, typecheck clean. Angad (whole repo, `npm run verify`): `astro check` 0 errors, **261/261 unit** ( `npm run test:unit`, all suites — CBAM's own share is 80/80, in `cbam-render.test.mjs` and `cbam-lines.test.mjs` ), publication contract 80 checks / 0 violations. The CBAM Playwright suite ( `tests/e2e/cbam-lines.spec.ts`, the only e2e file this branch touches) is

19/19 — up from 16 at the last count of this line, which predated both the 77-test unit file this branch grew to 80 and the 17-test e2e file this branch grew to 19. This line is now the number `npm run verify` and `npx playwright test tests/e2e/cbam-lines.spec.ts` print today, not a carried-forward figure from before either grew.

CBM's own `differential.test.ts` had pinned this gap at 382 priceable / 183 stranded for India and concluded it *"needs corpus research rather than a code change."* It was a code change. The pin now reads 528/37, and the remaining 37 are a genuine corpus question.

## 7. Open

| Item                           | Note                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSCF unpublished               | 94.3% of answers are what-ifs. Nothing we can do; a one-line data change when Brussels publishes.                                                                                                                                                                                                                                                                                                                         |
| 2027+ import dates             | Certificate prices exist only for 2026 quarters, so any later date fails closed with no explanation. The most visible remaining flaw.                                                                                                                                                                                                                                                                                     |
| Combined (C)/(F) routes        | 19 goods, resolution backed by both regulations. Not implemented.                                                                                                                                                                                                                                                                                                                                                         |
| Art 9 carbon price paid abroad | Not modelled; disclosed in output as making the figure conservative. Implementing act still a draft.                                                                                                                                                                                                                                                                                                                      |
| Art 2(2) citation              | Wrong in five places in the vendored engine — Article 2 has no numbered paragraphs; the provision is Art 1(2). Unreachable today. Our own new <code>cscf_status</code> CSV column, which prints this same fact for electricity's <code>zero_by_fiat</code> rows, cites it correctly as Art 1(2) — the two do not contradict each other; one is our export, the other is the still-unfixed vendored <code>sefa.ts</code> . |
| 41,100 default factors         | Never reconciled against IR 2025/2621's 2,400-page Annex I. The largest unaudited surface.                                                                                                                                                                                                                                                                                                                                |

Sources: IR (EU) 2025/2620 and 2025/2621 (OJ PDFs, hashes above); Reg (EU) 2023/956; the shipped rule pack at [deltaclimate.earth](https://deltaclimate.earth). Benchmark figures quoted are from our own pack, cross-checked against the published Annex.